Aarav Verma

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EDUCATION

University of Illinois at Urbana-Champaign
Grainger College of Engineering - Bachelor of Science in Aerospace Engineering with a Minor in Computer Science

WORK EXPERIENCE

Tarform - Champaign, IL Aug 2026 - Oct 2026
Mechanical Design Engineer Intern (Part-Time)

- Created and maintained production documentation, including manufacturing drawings, for machined and fabricated motorcycle components in SolidWorks, applying GD&T per ASME Y14.5 drafting standards
- Prepared manufacturing drawings and GD&T tolerance views for a back rack assembly (LH/RH side structures, rear structure, aluminum plate) in SolidWorks, checking fit and assembly across all components

LEADERSHIP & INVOLVEMENT

National Organization for Business and Engineering - Champaign, IL
Project Manager, Interceptor UAV June 2026 - Current

- Lead 6 developers across CAD and software tracks on a palm-sized interceptor UAV for Haylon Technologies, packaging a proprietary battery management system into a streamlined fuselage with four motor nacelles
- Review developer Fusion 360 assemblies for joint definition, interference, and center of mass, then integrate 7 features from a 4-way design competition into a final airframe on M3 heat-set inserts
- Direct development of a Python vision pipeline (YOLO, ByteTrack) that holds persistent tracks with EMA smoothing, ranges targets from a calibrated pinhole model, and solves an intercept quadratic for the lead point

Engineering Lead, Canary Drone Project Feb 2026 - May 2026

- Redesigned propeller to toroidal configuration in Fusion 360, simulating (ANSYS Fluent) improved thrust efficiency ~11% at 4-6k RPM and validating structural integrity via FEA (FoS > 2.0) under 12N thrust loads
- Designed a modular UAV airframe in Fusion 360 with a Jetson housing frame and tool-free foldable leg assembly, and presented the design at the Cozad New Venture Challenge against 300+ teams for $500,000

Ghost Electric Motorcycles - Champaign, IL Aug 2025 - Jan 2026
Drivetrain Engineer

- Engineered a load-bearing drivetrain frame and rear suspension swingarm in SolidWorks, reverse-engineering the 2010 Kawasaki ZX-6R to replicate OEM anti-squat ratio and pivot geometry within ±0.05 mm
- Validated drivetrain stiffness and load capacity in ANSYS FEA and MATLAB under peak torque delivery and chain tension across braking, cornering, and acceleration loading conditions

PROJECTS

Cantilever Bracket FEA - Olympia, WA Aug 2026

- Designed a hand-calc vs. FEA comparison experiment to validate tip deflection and bending stress predictions, catching a 10x error in the FEA load application by querying the true load-face area from the solved mesh
- Reconciled corrected FEA output to within 10% of hand calculations, attributing the remaining gap to bend-fillet compliance and mounting-hole stress concentrations the beam model omits

Parametric Airfoil Structural Analysis - Olympia, WA Aug 2026

- Generated NACA 0012 airfoil geometry in MATLAB, modeled the solid in Fusion 360, and imported STEP geometry into MATLAB for structural FEA under directional aerodynamic surface loads
- Designed and ran a mesh convergence experiment to test sensitivity of stress results to mesh refinement, found a stress singularity, and added a CAD fillet that brought successive mesh results within 2% of each other

Actuator-Limited Flight Controller - Champaign, IL Jul 2026 - Sep 2026
Simulink

- Designed a cascaded rate/angle PID controller (Betaflight/PX4-style) for a multi-axis UAV model, cutting peak overshoot from 40% to 10% by diagnosing and correcting a derivative-kick and noise-amplification problem
- Characterized actuator saturation and integral anti-windup behavior by sweeping the motor torque limit to locate the boundary between recoverable windup and an unrecoverable, underpowered-actuator failure mode
- Built a MATLAB 3D visualization pipeline that animates UAV altitude from logged Simulink flight data using composed rotation matrices, verifying transformation correctness through direct numerical checks

SKILLS

CAD & Design: Siemens NX, SolidWorks, Fusion 360, Onshape, GD&T (ASME Y14.5), 3D Printing (PLA, ABS)
Simulation & Analysis: ANSYS FEA, ANSYS Fluent (Aerodynamic CFD), MATLAB (PDE Toolbox), Simulink
Programming & Software: MATLAB, Python (NumPy, Matplotlib, Open CV, YOLO), Microsoft Office